

Exploratory Data Analysis (EDA) on Student Performance Dataset

1. Introduction

Exploratory Data Analysis (EDA) is the process of analyzing datasets to summarize their main characteristics using statistical methods and visualizations. The primary goal of this project is to explore student performance data, identify patterns and relationships, and derive meaningful insights.

This project uses the Student Performance Dataset and applies various EDA techniques such as descriptive statistics, correlation analysis, and data visualization.

2. Objective

The objectives of this project are:

- To understand the structure of the dataset.
- To analyze student performance across different subjects.
- To identify trends and correlations among variables.
- To visualize the data using charts and graphs.
- To derive meaningful insights from the dataset.

3. Dataset Description

Dataset Name:

Student Performance Dataset

Features Used:

- Math Score
- Reading Score
- Writing Score

Dataset Type:

Educational Data

Number of Records:

1000 (or the actual number in your dataset)

Number of Features:

3 numerical features used for analysis.

4. Tools and Technologies Used

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn

Development Environment

- Visual Studio Code (VS Code)

5. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the dataset using Pandas.
2. Checked dataset structure and information.
3. Verified data types of all columns.
4. Checked for missing values.
5. Generated descriptive statistics for numerical columns.

Result:

No missing values were found in the dataset.

6. Exploratory Data Analysis

6.1 Descriptive Statistics

Descriptive statistics were calculated to understand:

- Mean
- Median
- Standard Deviation

- Minimum Values
- Maximum Values

These statistics provide an overview of student performance across subjects.

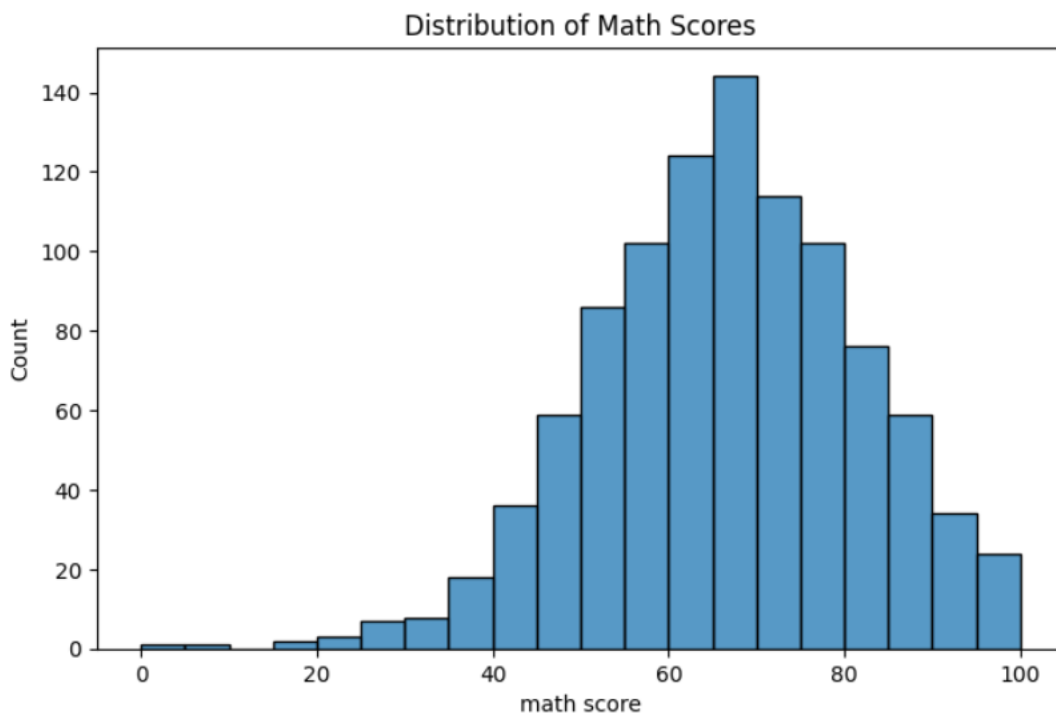
6.2 Distribution of Mathematics Scores

A histogram was created to visualize the distribution of mathematics scores.

Observation:

- Most students scored between 50 and 80.
- The distribution appears approximately normal.
- Very few students scored extremely low or extremely high marks.

Histogram :



6.3 Correlation Analysis

Correlation analysis was performed to identify relationships among:

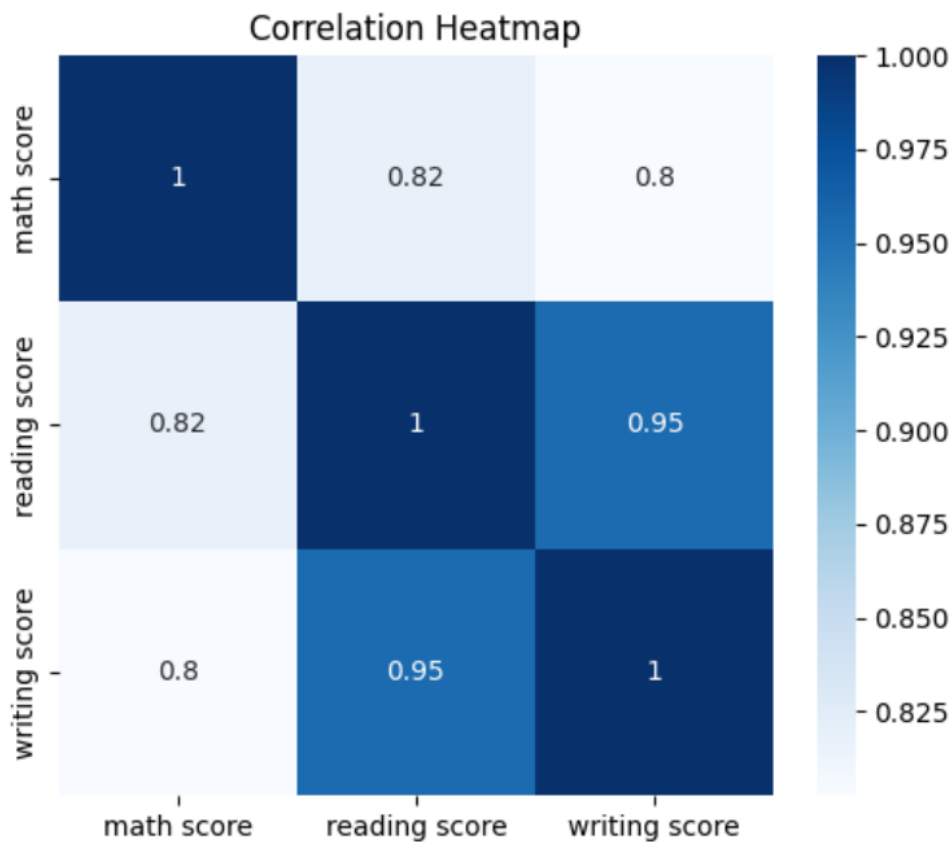
- Math Score
- Reading Score
- Writing Score

A correlation matrix was generated and visualized using a heatmap.

Observation:

- Reading and Writing scores show a strong positive correlation.
- Students who perform well in Reading generally perform well in Writing.
- Mathematics scores also have a positive relationship with Reading and Writing scores.

Correlation Heatmap :



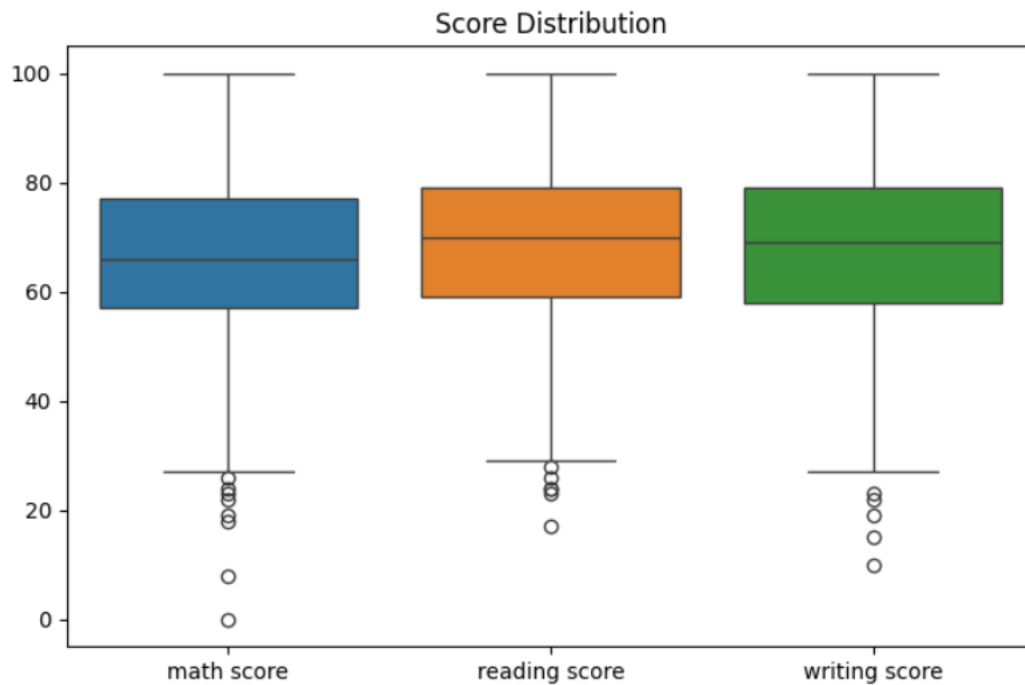
6.4 Outlier Analysis

A box plot was used to identify potential outliers in the dataset.

Observation:

- A few outliers are present.
- Most students have scores within a consistent range.
- No significant abnormal data points were detected.

Box Plot:



7. Key Findings

The following insights were obtained from the analysis:

1. Most students scored between 50 and 80 marks.
2. Reading and Writing scores are highly correlated.
3. Student performance is generally consistent across subjects.
4. Very few extreme outliers exist in the dataset.
5. Reading and Writing scores tend to increase together.

8. Challenges Faced

- Understanding relationships between variables.
- Selecting appropriate visualizations for analysis.
- Interpreting statistical summaries and correlation values.

9. Conclusion

This Exploratory Data Analysis project successfully analyzed student performance data and identified important trends and relationships. Statistical summaries and visualizations helped in understanding score distributions, correlations, and potential outliers.

The analysis revealed a strong positive relationship between Reading and Writing scores and showed that most students performed within a moderate score range. This project provided practical experience in data exploration, visualization, and analytical thinking using Python.

10. Future Scope

Future improvements may include:

- Analyzing additional student attributes.
- Building predictive machine learning models.
- Applying advanced visualization techniques.
- Creating interactive dashboards using Power BI or Tableau.

Author

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Thiranex Internship – Task 3

Exploratory Data Analysis (EDA) Project